A CASE-STUDY BASED REPORT

ON

Machine Learning Based Wine Quality Prediction Using Decision Trees and Random Forest

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INDEX

Abstract

Introduction .....

Background .....
Motivation for the Project .....
Overview of the Work Carried Out.....
Scope and Responsible-Use Statement.....

Problem Statement .....

Case-Study Scenario.....
Central Case-Study Question .....
Formal Problem Definition .....
Objectives that Guide the Study.....
Assumptions and Limitations .....

Dataset and Exploratory Data Analysis

About the Dataset .....
Purpose of EDA.....
Preparation Steps Actually Performed.....
EDA Exploration - Implementation.....
Summary of EDA Findings.....

Algorithm Selection and Implementation .....

Conclusion

Signature of the Team members

Abstract

Machine Learning has become useful for analysing data and making predictions in real-world applications. This case study, titled “Wine Quality Prediction Using Decision Trees and Random Forest,” investigates how the physicochemical properties of red wine can be used to predict its quality category. The project uses the red wine quality dataset described in the project materials. The presentation identifies 1,599 original samples and 11 input features: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. It reports that 240 duplicate records were removed, leaving 1,359 records. The wine quality score is grouped into Poor (score 4 or below), Average (score 5–6) and Good (score 7 or above). Decision Tree and Random Forest classifiers are compared. The workflow described in the abstract includes data preparation, train/test splitting, hyperparameter tuning using GridSearchCV, cross-validation and evaluation using accuracy, precision, recall, F1-score, confusion matrix, ROC curve and AUC. The WineSense AI website provides Home, Predict, Dataset and Results pages. The review presentation reports accuracy of 84.6%, precision of 79.7%, recall of 84.6% and AUC of 0.887; these values must be verified against the final experiment before being treated as confirmed results. The system is intended to support quality assessment, not replace expert sensory evaluation.

1. Introduction

1.1 Background

Wine quality influences customer satisfaction, market value and product reputation. Traditionally, experienced evaluators assess wine through sensory characteristics such as taste, aroma, colour and texture. Although expert evaluation is valuable, it can be time-consuming and may vary between evaluators. Machine Learning can learn patterns from previously labelled examples and use measurable chemical properties to estimate the quality category of a new sample. In this project, the chemical measurements are the input features and the recorded wine quality score is the target.

1.2 Motivation for the Project

The motivation is to explore a consistent and data-supported way of estimating wine quality. Automated predictions may help reduce repeated manual assessment and support quality-control workflows. The project also provides a comparison between a Decision Tree, which is relatively easy to interpret, and Random Forest, which combines several decision trees. Comparing both algorithms helps identify which model performs better under the selected evaluation procedure.

1.3 Overview of the Work Carried Out

The project materials describe the following workflow: load the wine dataset; inspect and prepare the data; select the 11 physicochemical input features; split the data into training and testing sets; train Decision Tree and Random Forest classifiers; tune hyperparameters using GridSearchCV and cross-validation; evaluate the models with classification metrics and visualisations; and provide a website interface for prediction and data exploration. The supplied website screenshots show a home page, prediction dashboard, dataset explorer and results dashboard. The report distinguishes features described in the abstract from numerical findings that need verification in the final code and experiment.

1.4 Scope and Responsible-Use Statement

The scope is to estimate a broad wine quality category from 11 physicochemical measurements using supervised classification. The result is a prediction based on patterns in the dataset, not a guarantee of taste, safety, authenticity or consumer preference. The system should be treated as a decision-support aid. It should not replace laboratory testing, regulatory checks or trained sensory evaluation. Predictions should be interpreted with the dataset's limitations in mind, and the model should be tested on suitable unseen data before being used in an operational setting.

2. Problem Statement

2.1 Case-Study Scenario

Consider a winery or quality-control laboratory that needs to assess many red wine samples. Sensory evaluation remains important, but reviewing every sample manually can require time and may lead to differences between evaluators. The team wants to investigate whether recorded chemical measurements can be used to make an initial quality estimate. This case study uses the Wine Quality dataset to compare Decision Tree and Random Forest as possible prediction methods.

2.2 Central Case-Study Question

Can a supervised Machine Learning model use 11 physicochemical properties to classify red wine into Poor, Average or Good quality categories, and how does Random Forest compare with Decision Tree when evaluated using appropriate classification metrics?

2.3 Formal Problem Definition

Let each wine sample be represented by a feature vector $X = (x1, x2, \ldots, x11)$, containing the 11 chemical measurements. Let y be the target class derived from the wine quality score: Poor if the score is 4 or lower, Average if the score is 5 or 6, and Good if the score is 7 or higher. The task is supervised multi-class classification. The model learns a function $f(X) \rightarrow y$ from labelled training examples and its generalisation is assessed using data kept separate from training.

2.4 Objectives that Guide the Study

Understand the red wine quality dataset and the meaning of its physicochemical features.

- The dataset's recorded quality score is treated as the correct target label for training and evaluation.
- The quality classes follow the boundaries listed in the project presentation: Poor ≤ 4, Average = 5–6 and Good ≥ 7.
- The model is assumed to be used on samples whose measurements are represented reasonably well by the training dataset.
- The dataset may not represent all wine varieties, production methods, regions or vintages.
- Chemical measurements do not capture every sensory factor or individual preference.
- The presentation reports 240 duplicate rows removed, leaving 1,359 records; this should be confirmed against the actual preprocessing code.
- The review presentation reports one set of metrics, while the website Results screenshot displays metric cards and includes a notice that values should be connected to real model results. The final values and website integration must therefore be checked.

- EDA counts and correlation findings are not included as verified facts unless they are generated from the actual dataset.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project materials identify the UCI Wine Quality red wine dataset. The review presentation reports 1,599 original samples, 11 input features and 240 duplicate records removed, leaving 1,359 records. The dataset contains numerical measurements describing wine chemistry, along with a quality score used as the target. The target score is grouped into three classes for the website and classification task.

Input feature	Meaning in simple words
Fixed acidity	Non-volatile acids present in the wine.
Volatile acidity	Volatile acids that can affect the wine's flavour profile.
Citric acid	Amount of citric acid measured in the sample.
Residual sugar	Sugar remaining after fermentation.
Chlorides	Measurement related to chloride or salt content.
Free sulfur dioxide	Free SO_2 measured in the wine.
Total sulfur dioxide	Total SO_2 measured in the wine.
Density	Mass per unit volume; related to the wine's composition.
pH	A measure related to acidity.
Sulphates	Sulphate content measured in the sample.
Alcohol	Alcohol content of the wine.

Target classes used in the project:

- Poor: recorded quality score ≤ 4.
- Average: recorded quality score 5–6.
- Good: recorded quality score ≥ 7.

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is performed before model training to understand the structure and quality of the data. It helps identify the number of rows and columns, data types, missing values, duplicate records, feature distributions, unusual values, target-class balance and relationships between features. EDA supports informed preprocessing and helps avoid drawing conclusions from data that have not been inspected.

3.3 Preparation Steps Actually Performed

The project abstract describes handling missing values, removing duplicate records, selecting relevant features and splitting the data into training and testing sets. The presentation specifically reports 240

duplicate records removed from 1,599 original samples, leaving 1,359 records. The provided materials do not include the complete preprocessing code or a run log, so the duplicate count is a reported project value and should be checked against the code before final submission.

Load the dataset into a Pandas DataFrame.

Inspect the row count, column names and data types.

Check each column for missing values and record the handling method.

Count duplicate rows and verify the reported removal of 240 records.

Confirm that 1,359 rows remain after the stated duplicate-removal step.

Select the 11 physicochemical features and the quality-score target.

Map quality scores to Poor, Average and Good using the documented boundaries.

Split the data into training and testing sets and record the split ratio and random seed used.

3.4 EDA Exploration – Implementation

The website's Dataset page displays all 1,599 original red-wine samples, chemical measurements, actual quality values and filters. Its summary cards show 1,599 samples, a 92.6% Random Forest match rate across all displayed rows, a 92.3% match rate for unique rows and 240 duplicate rows flagged. These are website-displayed dataset summaries; the match rates should not be described as the same as the held-out test accuracy unless the calculation method and data scope are confirmed.

The following EDA checks should be included only when the corresponding outputs have been generated from the project dataset:

- Dataset shape and data types: report the actual number of rows and columns.
- Missing-value summary: list the actual missing count for each feature.
- Duplicate analysis: report duplicate counts before and after cleaning.
- Quality-score distribution: create a bar chart of the original target scores.
- Class distribution: create a bar chart for Poor, Average and Good.
- Feature distributions: use histograms or boxplots for alcohol, volatile acidity, density and sulphates.
- Correlation heatmap: show pairwise correlations among numerical variables.
- Feature-versus-class comparisons: inspect whether feature distributions differ across target classes.

Insert the generated plots with captions and briefly explain what each plot actually shows. Do not invent missing-value counts, correlations or class totals when the source data or output is not available.

3.5 Summary of EDA Findings

From the supplied project materials, the confirmed reported dataset summary is 1,599 original samples, 11 input features and 240 duplicate records flagged or removed, with 1,359 rows stated to remain after cleaning. The website displays 92.6% match rate across all rows and 92.3% for unique rows; these appear to describe agreement between displayed predictions and actual labels in the dataset explorer, not necessarily the independent test-set performance. The materials do not provide verified missing-value counts, a correlation matrix or target-class counts. Those findings should be added after running the EDA code.

4. Algorithm Selection and Implementation

4.1 Algorithm Selection

Decision Tree and Random Forest are supervised learning classifiers. They are suitable for this case study because the input consists of numerical features and the output is a known category. The project compares them on the same classification task to evaluate their performance.

4.2 Decision Tree

A Decision Tree makes a prediction by applying a sequence of feature-based rules. Each split asks a question about a feature, branches represent possible outcomes, and a leaf gives the predicted class. The method is relatively easy to explain. However, if the tree becomes too complex, it may overfit the training data and perform less well on new samples.

4.3 Random Forest

Random Forest combines multiple decision trees. Each tree is trained using variation in the data and features, and the trees' outputs are combined to produce a final prediction. This approach often reduces the instability of a single tree and can improve generalisation. Its actual advantage must still be demonstrated through fair evaluation on held-out data.

4.4 Implementation Workflow

Load and inspect the dataset.

Prepare the records and select the 11 input features and target class.

Separate the training data from the held-out testing data.

Train a Decision Tree classifier.

Train a Random Forest classifier.

Use GridSearchCV and cross-validation to compare hyperparameter settings, as described in the abstract.

Evaluate the selected models on the test data.

Generate evaluation outputs such as accuracy, precision, recall, F1-score, confusion matrix, ROC curve and AUC where supported.

Save/load the trained model with Joblib if this step is implemented.

Connect the verified model to the prediction interface and check that the displayed output is generated by that model.

4.5 Evaluation Metrics

- Accuracy: the proportion of all predictions that are correct.
- Precision: among the samples predicted as a class, the proportion that truly belongs to that class.
- Recall: among all samples truly belonging to a class, the proportion correctly identified.
- F1-score: combines precision and recall into one measure.
- Confusion matrix: shows how many samples from each actual class were assigned to each predicted class.
- ROC curve and AUC: evaluate how well the model separates classes across thresholds; for multiclass tasks, the averaging approach should be stated.

- Cross-validation: repeats training/validation across multiple folds to examine consistency.

4.6 Results Reported in the Review PPT

The review PPT reports 84.6% accuracy, 79.7% precision, 84.6% recall and AUC 0.887. It lists these feature-importance values: Alcohol 0.161, Volatile acidity 0.133, Sulphates 0.132, Density 0.090 and Total sulfur dioxide 0.086. The PPT states that Random Forest performed better overall than Decision Tree. These values should be treated as results reported in the review presentation until the final evaluation output confirms the model, test split, metric averaging method and AUC calculation. A verified F1-score was not present in the supplied review deck.

4.7 Website Implementation and Demonstration

The WineSense AI website provides four screens supplied for this report. The Home page introduces the project and links to Predict, Dataset, Methodology, Models, Results and About. The Predict page provides editable fields for 11 chemical inputs, buttons to add samples, fill sample data and clear entries, and a table displaying predicted quality categories and confidence text. The Dataset page shows sample records and filtering controls. The Results page presents cards for accuracy, precision, recall, F1-score, AUC and five-fold cross-validation accuracy. Because the Results screenshot notes that values are placeholders until real evaluation values are connected, confirm the underlying implementation before treating those website metrics as final experimental results.

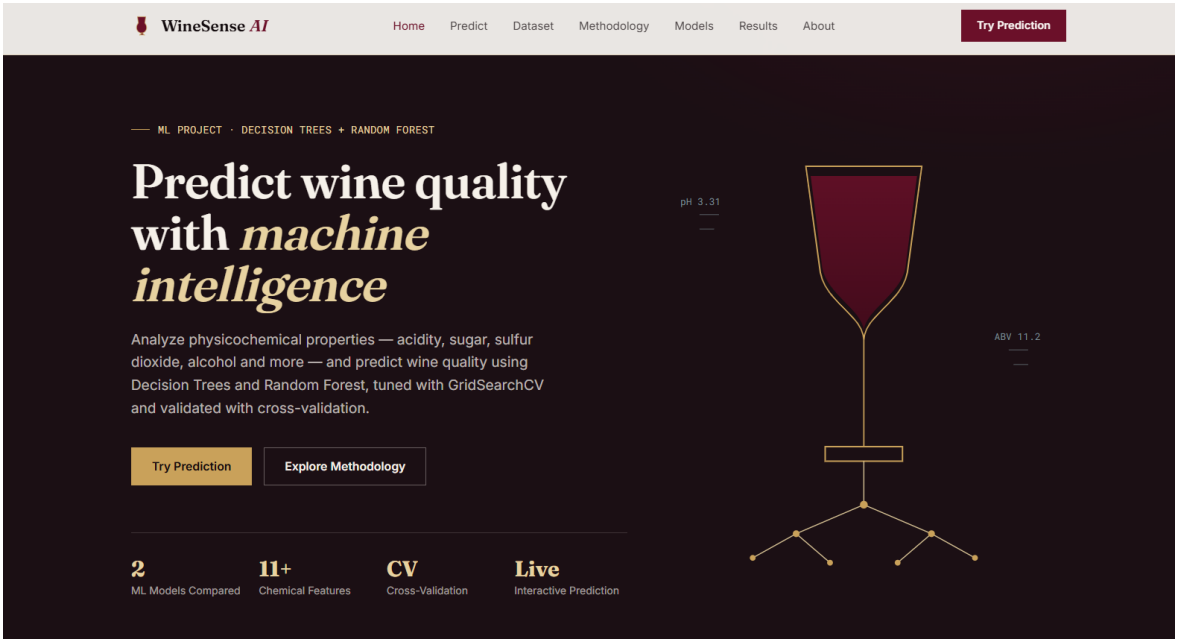


Figure 1. WineSense AI home page.

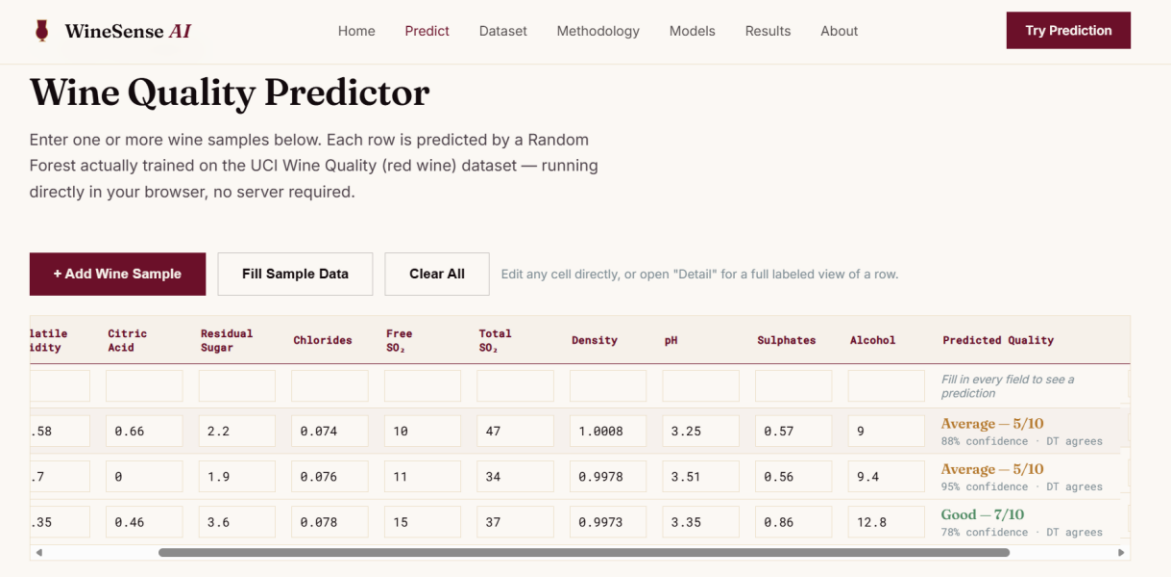


Figure 2. Wine quality prediction dashboard with sample inputs.

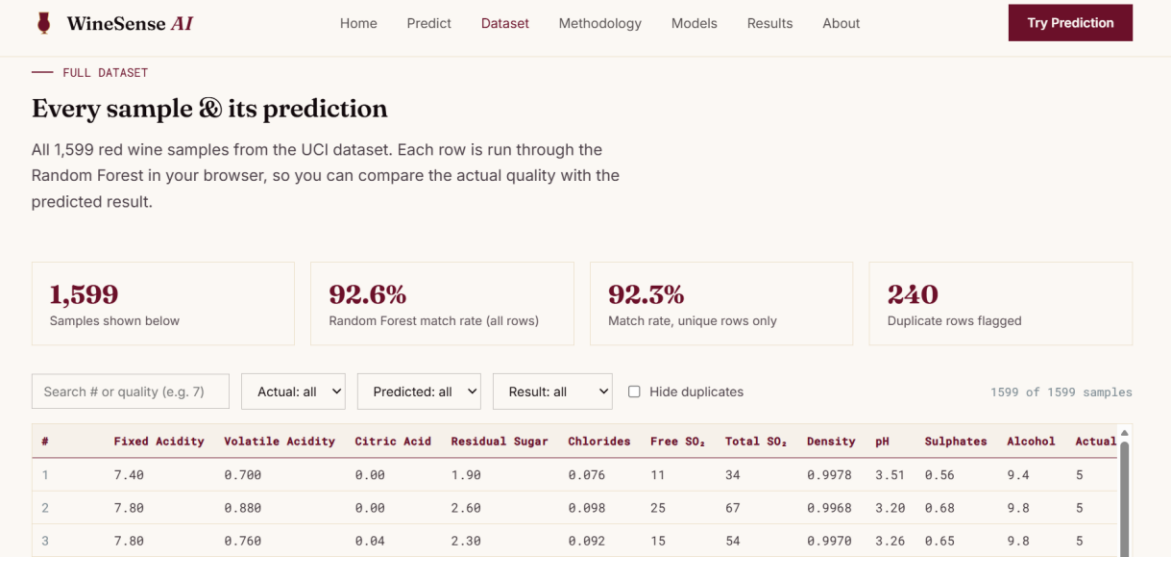


Figure 3. Dataset explorer showing red-wine samples and filters.

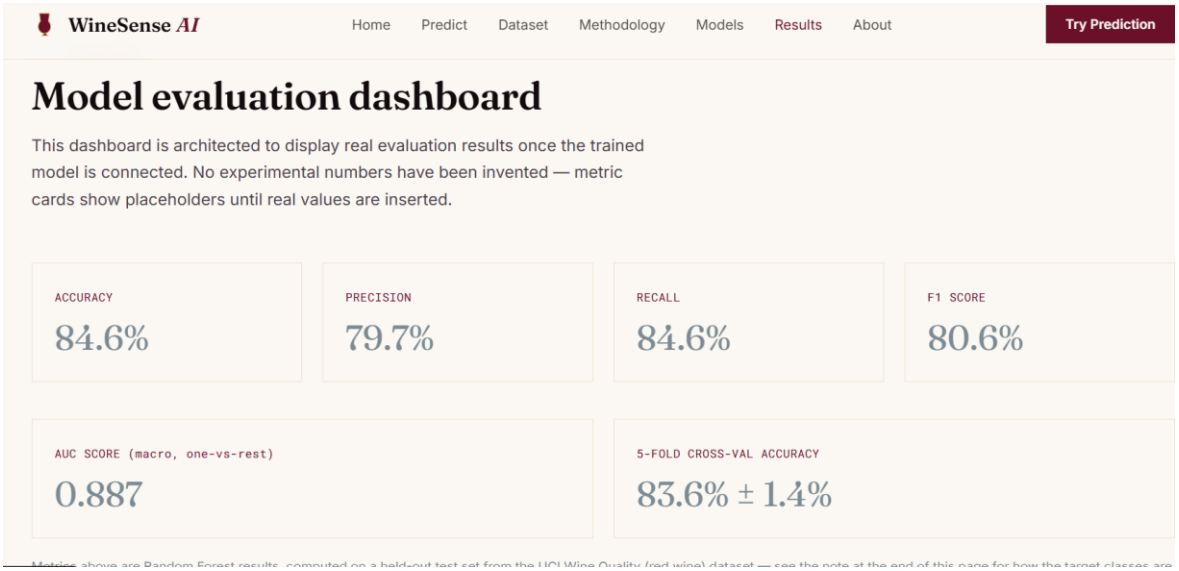


Figure 4. Model evaluation dashboard displayed by the website.

Suggested demonstration flow: (1) open Home and explain the purpose of the application; (2) open Predict, select Fill Sample Data, and point out the 11 input features and predicted categories; (3) open Dataset and show sample rows, filters and duplicate/match summaries; (4) open Results and explain the evaluation metrics, noting that their connection to the final trained model must be verified.

5. Conclusion

This case study investigates Machine Learning for predicting red wine quality from 11 physicochemical properties. The project compares Decision Tree and Random Forest classifiers and describes a workflow covering dataset preparation, training, hyperparameter tuning, evaluation and a website interface. The review PPT reports accuracy of 84.6%, precision of 79.7%, recall of 84.6% and AUC of 0.887, and states that Random Forest performs better overall. These results must be verified against the final experimental output before being presented as confirmed findings. The WineSense AI website demonstrates the intended user journey through Home, Predict, Dataset and Results pages. The system may support initial quality screening, but its predictions depend on the training data and do not replace expert sensory evaluation. Future improvements include testing on more diverse samples, evaluating additional algorithms, confirming full model-to-website integration and monitoring performance on new data.

References

1. UCI Machine Learning Repository. Wine Quality dataset (red wine). Confirm and include the exact dataset URL that corresponds to the dataset file used in the project.

2. Project abstract: “Wine Quality Prediction Using Decision Trees and Random Forest,” supplied by the project team.

3. Project review presentation: Wine Quality Prediction Using Decision Trees and Random Forest, supplied project PPT.

4. WineSense AI website screenshots supplied by the project team.

5. Scikit-learn documentation for DecisionTreeClassifier, RandomForestClassifier, GridSearchCV and classification metrics, if these components are used in the final implementation.

Final Verification Before Submission

- Confirm the team members' names and roll numbers and the faculty guide details.
- Insert actual EDA charts and replace any blank findings with values calculated from the dataset.
- Verify the reported 240 duplicate records and final row count from the code.
- Verify the train/test split, GridSearchCV settings and cross-validation procedure.
- Recalculate or confirm all final metrics, including F1-score and multiclass AUC settings.
- Confirm that website predictions and metric cards are connected to the final trained model.
- Add page numbers to the index after final pagination and obtain required team signatures.